



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 20

TITLE : Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Problem Statement

Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives

To perform database modification operations using JDBC.

Theory

JDBC allows Java applications to perform database manipulation using SQL Data Manipulation Language (DML) statements such as INSERT, UPDATE, and DELETE. The INSERT statement adds new records into database tables, UPDATE modifies existing records, and DELETE removes unwanted records. These operations are executed using the Statement or PreparedStatement interface. PreparedStatement is generally preferred because it supports parameterized queries, improves performance, and prevents SQL injection attacks. Proper transaction management ensures that database modifications are performed safely and consistently. These operations form the basis of almost every database-driven application such as banking systems, hospital management systems, and student information systems.

Output Screenshots

```
• dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp20/Main.java
• dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp20.Main
Record inserted successfully.
Record updated successfully.
Record deleted successfully.
❖ dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

CODE 1

```
1 package Exp20;
2
3 import java.sql.*;
4
5 public class Main {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/college2",
12                "root",
13                "devansh@2007"
14            );
15
16            Statement stmt = con.createStatement();
17
18            // INSERT
19            stmt.executeUpdate(
20                "INSERT INTO student VALUES (101, 'Rahul', 'Java',
21            85)"
22            );
23            System.out.println("Record inserted successfully.");
24
25            // UPDATE
26            stmt.executeUpdate(
27                "UPDATE student SET marks = 90 WHERE roll_no = 101"
28            );
29            System.out.println("Record updated successfully.");
30
31            // DELETE
32            stmt.executeUpdate(
```

```

32         "DELETE FROM student WHERE roll_no = 101"
33     );
34     System.out.println("Record deleted successfully.");
35
36     con.close();
37 }
38 catch (Exception e) {
39     System.out.println("Error: " + e.getMessage());
40 }
41 }
42 }

```

EXERCISE 1

```

1 package Exp20.Exercise;
2
3 import java.sql.*;
4
5 public class Exp1 {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/company",
12                "root",
13                "devansh@2007"
14            );
15
16            Statement stmt = con.createStatement();
17
18            // CREATE - Insert employee
19            stmt.executeUpdate(
20                "INSERT INTO employee VALUES (101, 'Rahul', 'IT',
21                50000)"
22            );
23            System.out.println("Employee inserted successfully.");
24
25            // UPDATE - Update salary
26            stmt.executeUpdate(

```

```

26         "UPDATE employee SET salary = 55000 WHERE emp_id =
101"
27     );
28     System.out.println("Employee updated successfully.");
29
30     // READ - Display employee
31     ResultSet rs = stmt.executeQuery(
32         "SELECT * FROM employee"
33     );
34
35     System.out.println("\nEmployee Records:");
36
37     while (rs.next()) {
38         System.out.println(
39             "ID: " + rs.getInt("emp_id") +
40             ", Name: " + rs.getString("name") +
41             ", Department: " + rs.getString("department") +
42             ", Salary: " + rs.getDouble("salary")
43         );
44     }
45
46     // DELETE - Delete employee
47     stmt.executeUpdate(
48         "DELETE FROM employee WHERE emp_id = 101"
49     );
50     System.out.println("Employee deleted successfully.");
51
52     con.close();
53 }
54 catch (Exception e) {
55     System.out.println("Error: " + e.getMessage());
56 }
57 }
58 }

```

EXERCISE 2

```
1 package Exp20.Exercise;
2
3 import java.sql.*;
4
5 public class Exp2 {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/college2",
12                "root",
13                "devansh@2007"
14            );
15
16            Statement stmt = con.createStatement();
17
18            // CREATE - Insert student
19            stmt.executeUpdate(
20                "INSERT INTO student VALUES (101, 'Amit', 'Java',
21                85)");
22            System.out.println("Student inserted successfully.");
23
24            // UPDATE - Update marks
25            stmt.executeUpdate(
26                "UPDATE student SET marks = 90 WHERE roll_no = 101"
27            );
28            System.out.println("Student updated successfully.");
29
30            // READ - Display students
31            ResultSet rs = stmt.executeQuery(
32                "SELECT * FROM student"
33            );
34
35            System.out.println("\nStudent Records:");
36
37            while (rs.next()) {
38                System.out.println(
```

```

39         "Roll No: " + rs.getInt("roll_no") +
40         ", Name: " + rs.getString("name") +
41         ", Course: " + rs.getString("course") +
42         ", Marks: " + rs.getInt("marks")
43     );
44 }
45
46 // DELETE - Delete student
47 stmt.executeUpdate(
48     "DELETE FROM student WHERE roll_no = 101"
49 );
50 System.out.println("Student deleted successfully.");
51
52 con.close();
53 }
54 catch (Exception e) {
55     System.out.println("Error: " + e.getMessage());
56 }
57 }
58 }

```

SQL CODE:

```

1 CREATE DATABASE company2;
2
3 USE company2;
4
5 CREATE TABLE employee (
6     emp_id INT,
7     name VARCHAR(50),
8     department VARCHAR(50),
9     salary DOUBLE
10 );
11
12 CREATE TABLE student (
13     roll_no INT,
14     name VARCHAR(50),
15     course VARCHAR(50),
16     marks INT
17 );
18

```

```

19 CREATE DATABASE college2;
20 USE college2;
21
22 CREATE TABLE student (
23     roll_no INT,
24     name VARCHAR(50),
25     course VARCHAR(50),
26     marks INT
27 );

```

Exercise

Q1. Create a Java application to manage employee records using CRUD operations.

```

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp20/Exercise/Exp1.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp20.Exercise.Exp1
Employee inserted successfully.
Employee updated successfully.

Employee Records:
ID: 101, Name: Rahul, Department: IT, Salary: 55000.0
Employee deleted successfully.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$

```

Q2. Create a Java application to manage student records using CRUD operations. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

```

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp20/Exercise/Exp2.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp20.Exercise.Exp2
Student inserted successfully.
Student updated successfully.

Student Records:
Roll No: 101, Name: Amit, Course: Java, Marks: 90
Student deleted successfully.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$

```

Conclusion

Thus, the experiment successfully demonstrated database connectivity using JDBC and performing INSERT, UPDATE, and DELETE operations on database records. The use of SQL DML statements through JDBC was understood, along with the role of Statement and PreparedStatement in modifying database data. The experiment also provided an understanding of how CRUD operations are used to manage records in real-world database applications.